

Integration of IndoRoberta Financial-BiGRU for Bitcoin Price Prediction

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Abstract—Bitcoin’s extreme volatility makes accurate price forecasting difficult, necessitating a hybrid approach that combines historical data and market sentiment. This study aims to develop a Bitcoin price forecasting model by integrating Indo RoBERTa Financial for Indonesian-language news sentiment analysis and the Bidirectional Gated Recurrent Unit (BiGRU) for time-series modeling. Quantitative data includes the closing prices of Bitcoin, GLD, the S&P 500, and the NASDAQ 100, as well as trading volume for the period January 1, 2024–September 30, 2025, obtained from Investing.com. Text data consists of 8,818 news articles from CNBC Indonesia, Detik.com, and Tokocrypto, whose sentiment was analyzed using Indo RoBERTa Financial. The results show that the external factors model delivered the best performance with a MAPE of 1.90%, outperforming the technical model (2.02%) and the hybrid model (2.71%). These findings demonstrate that integrating historical Bitcoin data with external factor indicators yields more accurate predictions than approaches based solely on news sentiment. Local news sentiment is highly noisy and less relevant for short-term Bitcoin predictions. The best model predicts a Bitcoin price of \$113,723.70 on October 1, 2025, with practical implications as a reliable investment decision-making tool for both institutional and retail traders.

Keywords— BiGRU, Bitcoin, Deep Learning, GRU, Sentiment Analysis

I. INTRODUCTION (HEADING 1)

Trade systems continue to evolve over time, ranging from the barter system and the use of precious metals and banknotes to electronic money (*E-money*) [1]. However, the emergence of *E-money* still raises concerns about data privacy and potential misuse by third parties. As a solution to this problem, cryptocurrencies are technology-based, *blockchain-based*, and decentralized. Cryptocurrencies do not have a physical form like a currency but are only a block of data bound by a hash as its validation [2]. Bitcoin is the first cryptocurrency to be a pioneer in the crypto industry. Bitcoin serves as a peer-to-peer payment network that is fast and secure without requiring authorization from third parties, such as financial institutions, and is completely decentralized. The crypto industry is showing massive growth, especially in Indonesia throughout 2024, characterized by a surge in transaction value and a significant increase in the number of investors [3].

Although it offers very promising investment profit

potential, cryptocurrency asset trading carries a high risk of losses due to extreme price volatility [4], [5]. Bitcoin's price volatility is not only controlled by technical supply and demand factors, but it is also highly sensitive to market sentiment triggered by external dynamics. Macroeconomic conditions, global interest rate policies, and geopolitical tensions (such as conflicts between countries) have been shown to trigger sharp fluctuations in crypto market capitalization in a short period of time [6], [7]. Therefore, the gap between the potential profit and the high risk of loss is the basis for the importance of this study. This research aims to provide a more accurate and comprehensive analytical approach to support wiser investment decision-making, both for beginners and experienced investors.

Various conventional statistical methods such as ARIMA [8], [9] and ARCH-GARCH [10], [11] have previously been used to predict crypto prices, but they have limitations when faced with highly volatile data over long periods of time [12], [13]. To answer these challenges, this study proposes the development of a hybrid forecasting model that integrates two architectures *deep learning* up-to-date, namely Indo RoBERTa Financial and Bidirectional Gated Recurrent Unit (BiGRU). BiGRU is very reliable in modeling sequential or time series data such as historical prices because it can access future and past information from every point in the data sequence for a better understanding of the context [14], [15], [16], while Indo RoBERTa has the advantage of natural language processing (NLP) to quantify market sentiment from online news sources in Indonesian [17]. By combining historical data analysis and text sentiment, this hybrid model is expected to be able to produce more accurate Bitcoin price forecasts and *robust* to support wiser investment decision-making.

II. LITERATURE REVIEW

A. Bidirectional Gated Recurrent Unit (BiGRU)

Gated Recurrent Unit (GRU) was first introduced by [18]. GRU changes calculation *hidden layers* of the RNN, and adjust the three gates on the *Long Short-Term Memory* (LSTM) into two gates, namely *reset gates* and *update gates*. The GRU also incorporates *hidden states* to control the flow of text information at different times, so that the model can easily extract information from the previous text [19].

1) GRU Activation Function.

The GRU model has two activation functions, namely the sigmoid activation function (σ), and the tanh activation function which are described below.

a) Sigmoid Activation Function (σ)

The sigmoid function converts the input into a value between 0 and 1, which is used to control how much information from the past will be retained or forgotten. The sigmoid activation function is calculated using Equation 2.2.

$$\sigma(x) = \frac{1}{1+e^{-x}} \quad (2)$$

b) Tanh Activation Function

The tanh function produces an output between -1 and 1, and is used to generate a new representation of the current information to be combined with the previous memory. The tanh activation function is calculated using Equation 2.3.

$$\tau = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (3)$$

2) GRU Algorithm

GRU algorithm adjusts the three gates on the *Long Short-Term Memory* (LSTM) into two gates, namely *reset gates* and *update gates*. This design not only improves computing speed but also saves system memory and makes the training process simpler [11].

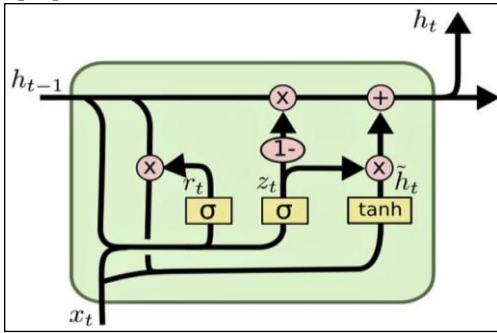


Fig. 1. GRU algorithm

Figure 1 shows the GRU algorithm obtained as follows.

$$r_t = \sigma(X_t W_{xr} + H_{t-1} W_{hr} + b_r) \quad (4)$$

$$z_t = \sigma(X_t W_{xz} + H_{t-1} W_{hz} + b_z) \quad (5)$$

$$\tilde{h}_t = t(X_t W_{xh} + r_t \odot H_{t-1} W_{hh} + b_h) \quad (6)$$

$$h_t = z_t \odot H_{t-1} + (1 - z_t) \odot \tilde{h}_t \quad (7)$$

where r_t , z_t , $\tilde{h}_t$, X_t , is the update gate, candidate hidden state, hidden state, and input time, respectively. Furthermore, W is the matrix corresponding to the parameter, b is the deviation parameter, and $\odot$ stands for element-wise product.

The GRU only considers previous information and does not consider the influence of later information on the current word, so it can lead to semantic bias [20], [21]. Therefore, the BiGRU bidirectional model was used in this study. The *Bidirectional Gated Recurrent Unit* (BiGRU) is a type of deep learning model that processes the sequence of data in two directions, namely forward and backward.

This allows the BiGRU model to access future and past information from every point in the data sequence for a better understanding of the context [22], [23]. The output of the BiGRU model at any given time is obtained by linearly calculating the hidden state information of the forward and backward GRU using the following equation.

$$\begin{cases} \vec{H}_t = GRU(x_t, H_{t-1}) \\ \overleftarrow{H}_t = GRU(x_t, H_{t+1}) \\ H_t = w_t \times \vec{H}_t + \vartheta_t \times \overleftarrow{H}_t + c_t \end{cases} \quad (8)$$

B. Text Analysis Sentiment using Indo RoBERTa Financial

Indonesian Robustly Optimized BERT Pretraining Approach (RoBERTa) *financial* is. optimization results of BERT, with improvements to the *pre-training* more efficient, such as the use of larger amounts of data, *batch size* higher levels, as well as task elimination *Next Sentence Prediction* (NSP). Using this optimization, IndoRoBERTa is able to better understand the context of the Indonesian language, thereby improving performance in tasks such as sentiment analysis, text classification, and other language modeling. In this study, the model used is ihsan31415/indo-roBERTa-financial-sentiment which has gone through an extensive pre-training process on financial news in Indonesian to understand the structure and meaning of language more accurately [24].

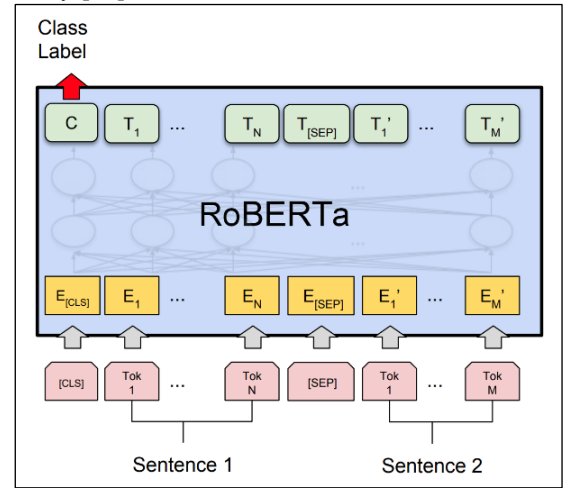


Fig. 2. RoBERTa Architecture

C. Model Performance Evaluation

Evaluation of model performance is one of the important aspects in building predictive models. The performance of the sentiment model in this study was measured using confusion matrix, while the forecasting model uses Mean Absolute Percentage Error (MAPE). The confusion matrix is an evaluation table used to measure the performance of classification models in machine learning by comparing the model's prediction to the actual label of the test data. MAPE presents the average error as a percentage of the actual value, making it easy to interpret by non-technical readers. The smaller the MAPE value, the better the quality of the model [25]. The values of accuracy and MAPE are formulated in equations (9) and (10), respectively.

$$accuracy = \frac{TT + TN}{TT + FT + TN + FN} \quad (9)$$

$$MAPE = \sum_{t=1}^n \frac{|A_t - F_t|}{A_t} \left(\frac{100\%}{n} \right) \quad (10)$$

Based on the research conducted by [26], the MAPE criteria are divided into four parts shown in Table 2.2.

TABLE I. MAPE CRITERIA

MAPE Values	Criteria
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$MAPE < 10\%$	Excellent
$10\% - 20\%$	Good
$20\% - 50\%$	Pretty good
$MAPE > 50\%$	Bad

Although MAPE is widely used because of its intuitive interpretation and scale-independent nature, it does not provide information regarding the magnitude of forecasting errors in the original units of the target variable nor the direction of prediction errors. For more comprehensive evaluation, we include additional metrics such as Root Mean Square Error (RMSE).

III. RESEARCH METHODOLOGY

A. Data Source

This study uses secondary data with a daily *time frame* obtained from investing.com websites and text data from 8,818 news articles on CNBC Indonesia, Detik.com, and Tokocrypto portals. News articles were taken using *web scraping* techniques with the keywords crypto, bitcoin, BTC, blockchain, cryptocurrency, geopolitics, trump tariffs, the fed as well as news on *Indonesian crypto exchange websites* such as Tokocrypto on the *bitcoin news*, *ethereum news*, and *policy regulation* pages during the period from January 1, 2024 to September 30, 2025.

B. Research Variables

The variables used in this study are Bitcoin Closing Price (X_1), GLD Price (X_2), S&P 500 (X_3), NASDAQ 100 (X_4), and Bitcoin Trading Volume (X_5). Text data for sentiment extraction were labeled as today's positive sentiment (D_1) and negative sentiment (D_2), as well as the previous day's (*lagged time*) positive sentiment (D_3) and negative sentiment (D_4).

When more than one news article was published on the same calendar day, the article-level sentiment scores generated by the *ihsan31415/indo-roBERTa-financial-sentiment* model were aggregated into a daily sentiment score. The aggregated score was subsequently converted into two binary sentiment indicators representing positive sentiment (D_1) and negative sentiment (D_2). In addition, D_3 and D_4 represent the one-day lagged values of D_1 and D_2 , respectively, to capture the effect of the previous day's sentiment on Bitcoin price movements. Days without published news articles were assigned a neutral value (0) for all sentiment indicators.

C. Analysis Steps

1. Collect historical data on Bitcoin (BTC) prices, stock market indices, including the S&P 500 and NASDAQ 100, and the gold exchange-traded fund (GLD) for the period from January 1, 2024, to September 30, 2025.
2. Collect Indonesian-language news articles based on predefined keywords using web-scraping techniques for the period from January 1, 2024, to September 30, 2025.
3. Preprocess the collected textual data to prepare it for sentiment analysis and model development.
4. Construct the final dataset by integrating all engineered features, including quantitative and textual features, into a single dataframe that is ready for modeling.

5. Divide the final dataset into training and testing sets, allocating 80% of the data for model training and 20% for model testing.
6. Conduct comparative modeling and training using two experimental scenarios to predict the target variable (Y_{t+1})
 - a. **Scenario 1:** Develop and train a standard BiGRU model using historical BTC price data (X_1) and BTC trading volume (X_5) as input variables.
 - b. **Scenario 2:** Develop and train a standard BiGRU model using all quantitative features (X_1) until (X_5) as input variables.
7. Develop and train the hybrid Indo-roBERTa-Financial-BiGRU model using all quantitative and textual features contained in the final dataset to predict the target variable (Y_{t+1}).
8. Evaluate the predictive performance of each model using the Mean Absolute Percentage Error (MAPE).
9. Select the best-performing model based on the lowest MAPE value.

IV. DRAW CONCLUSIONS AND FORMULATE RECOMMENDATIONS BASED ON THE OVERALL ANALYSIS RESULTS.RESULTS

A. Results of Cryptocurrency Market Sentiment Labeling Extraction Based on News Articles

The results of labeling news article text data using the Indo RoBERTa Financial model are shown in Table 2.

TABLE II. NEWS SENTIMENT LABELS

Label Sentimen	Number of Articles
Positive	1363
Negatives	2430
Neutral	5026

Table 2 shows the results of the labeling of cryptocurrency news sentiment using the Indo RoBERTa model Financial is unbalanced between sentiment categories, with the dominance of news labeled neutral as many as 5,026 articles, then negative news as many as 2,430 articles, and positive news as many as 1,363 articles. This proportion illustrates the characteristics of the *cryptocurrency* market, which tends to be influenced by media uncertainty and caution in reporting crypto asset movements. This condition reflects the characteristics of local news (Detik.com, CNBC, and Tokonews), which are more informative and less emotionally biased, such as daily price predictions by certain figures or institutions. Meanwhile, the almost double the negative news indicates the media's tendency to highlight extreme risks, regulations, or volatility in the crypto market, in line with the public's still skeptical perception of digital assets.

To evaluate the reliability of the sentiment-labeling process, manual validation was conducted on a randomly selected sample comprising 5% of the total dataset or around 440 of 8819 news articles collected from all news sources. The sentiment labels generated by the *ihsan31415/indo-roBERTa-financial-sentiment* model were compared against manually assigned labels. This manual comparison represents the full extent of external validation performed for the sentiment classifier in this study. Furthermore, to find out how accurate the sentiment model is, a confusion matrix

analysis will be carried out using 5% of the total data or around 440 articles in Figure 3.

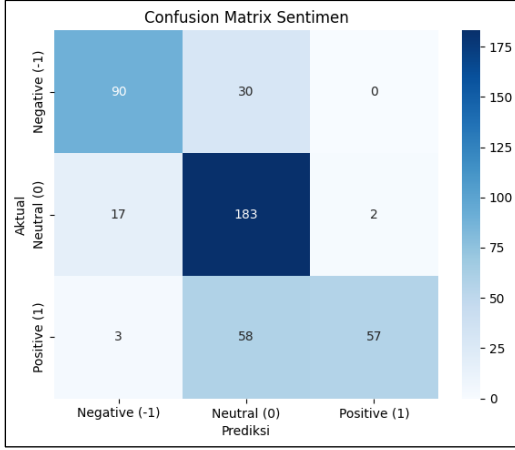


Fig. 3. Confusion Matrix of Sentiment Labeling

Figure 3 shows that the majority of cryptocurrency market sentiment analyzed using RoBERTa Financial's Indo model is neutral sentiment. This is reinforced by the results of manual labeling that categorize many neutral sentiments, resulting in a sentiment accuracy of 75%. Dominant neutral news may lack strong predictive signals that can affect the model's accuracy.

B. Bitcoin Price Modeling Using Bidirectional Gated Recurrent Unit (BiGRU)

1) First Scenario Model

The results of BiGRU modeling using 2 features namely historical of Bitcoin price and Bitcoin trading volume are obtained by hyperparameter tuning. The best combination of parameters in the form of 30% dropout, 32% batch size, max epoch 200, and early stopping with patience 10 resulted in the best performance, with training stopping in the 29th epoch. Early stopping allows the model to end training early when there is no significant increase in validation loss for 10 consecutive epochs, preventing wasted computing time while avoiding overfitting.

Moreover, the dropout value of 30% is also important because it deactivates some neurons randomly during training, so the model doesn't rely too much on specific patterns that may only appear occasionally on highly volatile crypto market data. Then, time step 30 shows that the model considers 30 days of recent historical data (price and volume) as input to generate a 1-day-ahead price forecast. The MAPE yield of 2.02% proves that the basic BiGRU can capture the dynamics of the Bitcoin market very well despite relying on only two simple features.

2) Second Scenario Model

The results of BiGRU modeling using five features, including Bitcoin historical price, GLD price, S&P 500 (X_3), NASDAQ 100, and Bitcoin trading volume, were obtained by the best combination of hyperparameters. The parameter combination of 30% dropout, a batch size of 32, a maximum of 200 epochs, and early stopping with a patience of 10 yielded the best performance, with training stopping at the 16th epoch. Early stopping allows the model to end training early when there is no significant improvement in validation loss for 10 consecutive epochs, preventing waste of computational time while avoiding overfitting.

Furthermore, a dropout rate of 30% also plays a crucial role because it randomly deactivates some neurons during training, ensuring the model does not become overly reliant on specific patterns that may only appear sporadically in highly volatile crypto market data. Furthermore, a time step of 30 indicates that the model considers the most recent 30 days of historical data (price and volume) as input to generate a 1-day-ahead price forecast. The MAPE result of 1.90% shows that adding these three external factor features enhances BiGRU's ability to understand the broader market context, thereby improving accuracy compared to the previous model, which used only two features.

C. Bitcoin Price Modeling using IndoRoberta Financial-BiGRU

At the peak of the study, hybrid modeling was carried out that integrated technical analysis, external factors, and market sentiment. The results show that the best combination of parameters in the form of 30% dropout, batch size 32, max epoch 200, and early stopping with patience 10 resulted in the best performance with the training stop in the 16th epoch. The use of early stopping allows the model to end the training process early when there is no significant increase in validation loss for 10 consecutive epochs, thus preventing wasting computing time while avoiding overfitting. The dropout value of 30% is also important because it deactivates some neurons randomly during training, so the model doesn't rely too much on specific patterns that may only appear occasionally on highly volatile crypto market data. Then, time step 30 shows that the model considers 30 days of recent historical data (price and volume) as input to generate a 1-day-ahead price forecast. The MAPE result of 2.71% indicates that the news sentiment data (D1 - D4) may contain significant noise, thereby reducing the advantages provided by historical data and external factors.

D. Best Model Bidirectional Gated Recurrent Unit (BiGRU)

The best BiGRU model after various modeling scenarios using a combination of various features is shown in Table 4.8.

TABLE III. PEROFMANCE COMPARISON OF THREE MODELS

Model	Dropout Rate	Batch Size	Epoch	Time Step	MAPE	RMSE
Technical Features	0.3	32	29	30	2.02%	1885.0
External Features	0.3	32	48	30	1.90%	1810.5
Hybrid Features	0.3	32	23	30	2.71%	2483.0

Table 3 shows that the best model overall is the external factor model with a MAPE value of 1.90% and the lowest RMSE. This model uses a combination of five predictive features, namely the historical price of Bitcoin (X_1), Bitcoin trading volume (X_5), the price of gold (X_2), the S&P 500 index (X_3), and the NASDAQ 100 index (X_4). This model, with a combination of 5 predictor variables, is slightly superior to the technical model that only uses 2 features (historical price and volume), with a MAPE of 2.02%. This confirms investors' assumption that Bitcoin's price movements are not only dependent on internal factors (price and volume) but also have a significant dependency on traditional financial market conditions and safe-haven assets

such as gold.

The Hybrid model, with the addition of the market sentiment feature (D_1 – D_4), showed a decrease in accuracy, with a MAPE of 2.71%, higher than the other two models. These results indicate that, in the context of this study, integrating news sentiment features in the format and extraction method used has not provided a strong enough signal to improve predictions, which may be caused by noise in text data or lag between news publications and slow market reactions. Next, a feature importance analysis will be carried out to find out which features have the greatest influence on Figure 4.

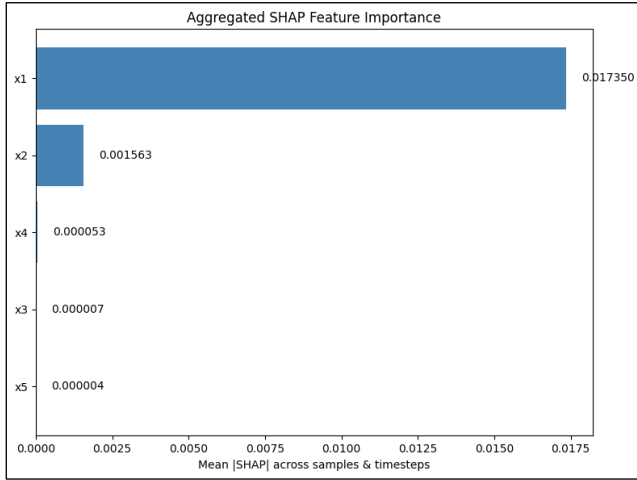


Fig. 4. Feature Importance

Figure 4 shows the relative contribution of each feature in the external factor model that combines historical data of Bitcoin (X_1), gold price (X_2), S&P 500 index (X_3), NASDAQ 100 index (X_4), and trading volume (X_5) to forecast the price of Bitcoin 1 day ahead. The results of SHAP's analysis reveal that the X_1 feature (Bitcoin's historical price) has the highest importance with the Mean |SHAP| of 0.017350, followed by the price of GLD with a value of 0.001563 (X_2). The price of GLD (X_2) is the second most important feature, indicating that *this safe-haven* asset has a significant influence on the dynamics of the Bitcoin market, especially during periods of economic uncertainty.

Additionally, a graph of the results of *the external factor model backtesting analysis* will be shown in Figure 5.

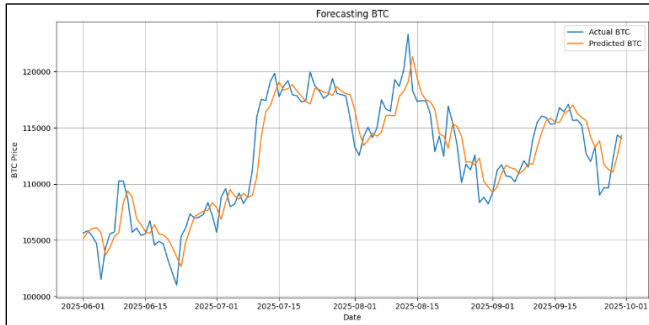


Fig. 5. Backtesting plot of the best model

Figure 5 shows the match of the Bitcoin price prediction line (orange) with the actual price (blue), especially in capturing price volatility patterns. Figure 4.9 also shows that BiGRU modeling has managed to identify various significant

movements, such as a sharp increase from early August 2025 to mid-August 2025, which immediately decreases until early September 2025. The integration of three additional features (X_2 – X_4) allows the model to understand the relationship between Bitcoin and traditional financial markets, where the movement of stock indices (S&P 500 and NASDAQ) as well as the price of gold provides context for external factors that influence the price of Bitcoin. The results of the Bitcoin price forecast using an external factor model on October 1, 2025 are \$113,723.70.

Furthermore, a graph of the *loss function* analysis will be shown in Figure 6.

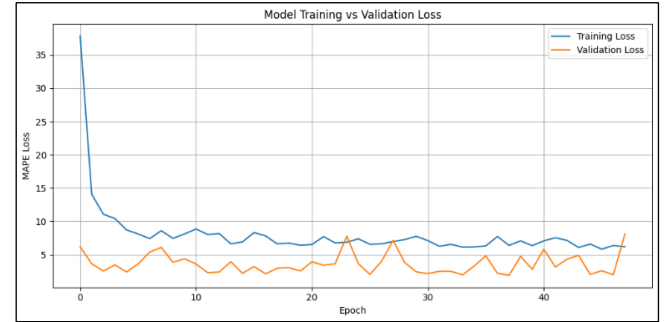


Fig. 6. Graphics Loss Function External Factors Model

Figure 6 shows a stable convergence pattern with MAPE as the main metric, where *the training loss* and *validation loss* decrease gradually from the beginning to the end. Although there were some fluctuations in *validation losses*, such as a significant increase in epoch 24 with a *validation loss* value of 7.7786, overall the model showed a positive trend with the lowest *validation loss* value of 1.90 reached in epoch 38, indicating an optimal point in training. The training process stops at epoch 48 earlier than the maximum of 200 epoch because *early stopping* with patience 10 is triggered. These results confirm that the BiGRU model is able to study temporal patterns well without overfitting, as can be seen from the consistency between *training loss* and *validation loss* that is not too different in the last epoch.

V. CONCLUSION

The integration of Bitcoin's historical data with external factor indicators (GLD, S&P 500, NASDAQ 100) provides a highly accurate prediction with a MAPE of 1.90%. The addition of local news sentiment actually lowered the model's performance due to the dominance of neutral news and high noise levels. Investors are advised to pay attention to the dynamics of macroeconomic indicators in addition to technical data when trading. Further research is recommended to use specific keywords related to institutional and regulatory fundamentals to improve the quality of news sentiment extraction.

Adding news sentiment features did not improve forecasting performance in this study. One possible explanation is the sentiment dataset's imbalance, dominated by neutral news articles, combined with a moderate sentiment-labelling accuracy of 75%. Consequently, the sentiment indicators may have introduced noise that reduced their contribution to the forecasting model. Furthermore, the binary sentiment representation may not fully capture the complex relationship between news dissemination and cryptocurrency market reactions.

Several limitations warrant consideration. Although this study reports both MAPE and RMSE, it does not include complementary metrics such as MAE, sMAPE, or directional accuracy, which could provide further insights into forecasting error, especially for highly volatile assets like Bitcoin. Furthermore, the authors did not compare the model with baseline statistical forecasting methods; the reported results are based on a single training run without formal significance testing of model differences. Consequently, the findings should be interpreted as comparative results within the evaluated model configurations, rather than as evidence of a definitive forecasting advantage.

Future research may repeat experiments with multiple random seeds; compare against statistical forecasting baselines; and conduct feature-level ablation studies to assess the individual contribution of sentiment variables. Moreover, using more targeted institution- and regulation-related keywords, such as Bitcoin ETF approvals, central bank policies, and cryptocurrency regulations, may improve the quality and relevance of sentiment information for forecasting.

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